

23-1 DHCP Configuration – Answer Key

In this lab you will perform a DHCP configuration for a small campus network. You will configure a router's outside interface as a DHCP client and then configure DHCP services using a Cisco router first and an external DHCP server later.

Cisco DHCP Client

1) Configure R1's outside interface as a DHCP client

Configure FastEthernet 0/0 on R1:

```
R1(config)#interface f0/0
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown
```

2) Verify that R1 received its public IP address

R1 should receive an address from the Service Provider. An example result is:

```
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned
DHCP address 203.0.113.2, mask 255.255.255.0, hostname R1
```

Verify with:

```
R1#show ip int brief
```

Expected example:

Interface Protocol	IP-Address	OK?	Method	Status
FastEthernet0/0 up	203.0.113.2	YES	DHCP	up
FastEthernet0/1 down	unassigned	YES	unset	administratively down
FastEthernet1/0 down	unassigned	YES	unset	administratively down
FastEthernet1/1 down	unassigned	YES	unset	administratively down
Vlan1 down	unassigned	YES	unset	administratively down

The exact DHCP-assigned address may vary.

3) What is the IP address of R1's DHCP server?

The DHCP server is:

```
203.0.113.1
```

View the DHCP lease information with:

```
R1#show dhcp lease
```

Example:

```
Temp IP addr: 203.0.113.2 for peer on Interface:
FastEthernet0/0
Temp sub net mask: 255.255.255.0
DHCP Lease server: 203.0.113.1 , state: Bound
DHCP Transaction id: 64B8EE07
Lease: 86400 secs, Renewal: 43200 secs, Rebind: 75600 secs
Temp default-gateway addr: 203.0.113.1
```

Cisco DHCP Server

4) Configure R1 as a DHCP server

The DHCP service must provide addresses to PCs in the 10.10.10.0/24 subnet.

Exclude 10.10.10.1 through 10.10.10.10 so these addresses remain available for servers and printers. The DNS server is 10.10.20.10.

Configure:

```
R1(config)#ip dhcp excluded-address 10.10.10.1 10.10.10.10
R1(config)#ip dhcp pool Flackbox
R1(dhcp-config)#default-router 10.10.10.1
R1(dhcp-config)#dns-server 10.10.20.10
R1(dhcp-config)#network 10.10.10.0 255.255.255.0
```

In this lab environment, enter the default-router and dns-server commands before the network command so clients receive all their IP settings correctly.

5) Configure the PCs as DHCP clients

In later versions of Packet Tracer:

1. Click the PC.
2. Open the **Desktop** tab.
3. Select **IP Configuration**.
4. Select **DHCP**.

6) Verify the clients received their IP information

Check the IP configuration on the PCs and confirm that they received their address, subnet mask, default gateway, and DNS information from DHCP.

7) Test DNS connectivity

From the PCs, ping the DNS server using its hostname:

```
DNSserver
```

DNS resolution may take some time.

8) Verify DHCP leases on R1

Verify that both clients received an IP address from R1's DHCP service.

9) Remove the Cisco DHCP configuration

Remove the DHCP configuration from R1 because the next part of the lab uses an external DHCP server:

```
R1(config)#no ip dhcp excluded-address 10.10.10.1 10.10.10.10  
R1(config)#no ip dhcp pool Flackbox
```

10) Release the PCs' DHCP addresses

On each PC, enter:

```
ipconfig /release
```

11) Verify DHCP is no longer available from R1

Enter:

```
ipconfig /renew
```

The PCs should no longer be able to obtain an IP address from R1 because the Cisco DHCP configuration has been removed.

External DHCP Server

12) Why are the PCs not receiving addresses from the external DHCP server?

The external DHCP server is at:

```
10.10.20.10
```

DHCP requests use **broadcast traffic**. Routers do not forward broadcast traffic by default, so R1 is not forwarding the DHCP requests from the PCs to the external DHCP server.

13) Configure R1 to forward DHCP requests

On the interface where the DHCP requests are received, configure R1 to forward the requests to the external DHCP server:

```
R1(config)#interface f0/1  
R1(config-if)#ip helper-address 10.10.20.10
```

The `ip helper-address` command allows R1 to relay DHCP requests to the external DHCP server.

14) Verify the clients received their IP information

Verify the PCs have successfully obtained their IP configuration from the external DHCP server.